

Krishnendu Karmakar

Senior Backend Engineer | Golang | Cloud & Distributed Systems

Kolkata, India | +91 8240903775 | krishnendukarmakar17@gmail.com | krish8learn.github.io

linkedin.com/in/krishnendu-karmakar-465687192 | medium.com/@krishkarmakar17

github.com/krish8learn | exercism.org/profiles/krish8learn | hackerrank.com/profile/krishkarmakar17

PROFESSIONAL SUMMARY

Senior Backend Engineer with 5+ years of experience designing and scaling distributed systems in Golang. At Kaleyra (TATA Communications), built high-throughput APIs and microservices processing 8,000+ RPS across multi-region AWS deployments, with deep expertise in gRPC, REST, GraphQL, RabbitMQ, AWS SQS, PostgreSQL, MongoDB, and Redis. Holds Executive PG Certification in Cloud Computing and DevOps from IIT Roorkee (AWS, Azure, Kubernetes, Docker, Terraform, CI/CD). Uses AI-assisted workflows (Cursor, Claude) to accelerate development and code review.

PROFESSIONAL EXPERIENCE

Kaleyra (TATA Communications Ltd.)

Senior Software Engineer · Jun 2022 – Present · Remote, India

Promoted from Software Engineering Professional to Senior Software Engineer

High-Throughput SMS Delivery Platform

- Architected and optimised a global SMS delivery system processing 8,000+ RPS for enterprise clients across India, Singapore, EU, and US, sustaining 50M+ requests/day at peak load.
- Built and maintained 10+ public REST APIs and 10+ internal gRPC APIs with strict validation, backward compatibility, and comprehensive error handling.
- Migrated RabbitMQ and Redis workers from PHP to Golang using Go's concurrency models (Worker pool & Fan-out) — cut message processing latency by ~80–90% and significantly reduced infrastructure load.
- Implemented priority-based message processing by SMS route type (OTP, transactional, marketing), with elevated priority handling for premium customer traffic to ensure faster delivery for time-sensitive messages.

Active-Active Disaster Recovery — Multi-Region AWS Architecture

- Contributed to dual-region AWS deployments across India (ap-south-1/2) and Southeast Asia (ap-southeast-1, ap-east-1), ensuring high availability and geographic redundancy with live traffic across all sites.
- Engineered a custom Redis Cache Moderator — a coordination layer maintaining Redis data consistency across geographically distributed deployments without Redis Enterprise, eliminating vendor dependency.
- Collaborated with database teams to implement bi-directional synchronisation across MySQL and MongoDB clusters ensuring consistency without service interruption.

Cross-Team API Integration

- Led integration of 5 cross-team APIs (Auth, Catalogue, Gateway, Billing, Callback) into the SMS platform for validations, data transitions, and balance deductions — coordinating across multiple internal teams.
- Implemented retry and fallback handling for API failures, ensuring SMS billing and validation pipelines remained operational during upstream outages.

Two-Way SMS Configurable System

- Coded and contributed to the architecture of a configurable two-way messaging system supporting automated replies and dynamic workflows, with zero-downtime deployments operating at 1000 RPS in India, Singapore, EU and US.
- Developed 5 internal CRUD REST APIs and 5 RabbitMQ workers orchestrating emails, SMS, webhooks, and subscriptions based on configurable customer keyword.

Multi-Channel OTP Verification System

- Built a fault-tolerant OTP service across SMS, Email, and Voice Call with retry and timeout mechanisms at every delivery layer.
- Replaced Redis with DynamoDB as primary state store for OTP verification, achieving eventual consistency and eliminating cache-layer failures under cross-region load.
- Integrated AWS SQS as a buffering layer between OTP generation and delivery services, improving throughput and isolating failure domains.

Engineering Practices & Platform Reliability

- Established structured logging (Uber Zap) and integrated New Relic for APM and event monitoring, alongside real-time request tracking — reducing average debugging time by ~60%.

- Implemented AES-256 encryption/decryption and data masking for PII, ensuring data-protection compliance.
- Built an API-key-based authorization middleware for request-level access control.

Wipro Limited

Software Engineer · Dec 2020 – May 2022 · Remote, Bengaluru

Computer Assembly Management Platform — HP (Client Project)

Inventory & Component Selection Application

- Built a platform for HP teams to view real-time component inventory (CPUs, monitors, batteries, keyboards) across factories in India, China, and Australia, supporting component selection and PDF-based processing requests routed via email.
- Developed 5 GraphQL APIs in Golang for frontend-backend communication, powering real-time inventory queries and component selection workflows.

Excel-to-PostgreSQL Data Import CLI

- Worked on a Golang CLI tool to import and validate Excel datasets into PostgreSQL, eliminating ~70% of manual data entry effort, with 80%+ unit test coverage reducing production regressions.

SKILLS & TECHNOLOGIES

Languages	Golang (primary) · Python · Rust (learning) · PHP
APIs & Protocols	REST · gRPC · GraphQL · WebSockets
Databases	PostgreSQL · MySQL · MongoDB · Redis · DynamoDB
Message Brokers	RabbitMQ · AWS SQS · Redis Queue
Cloud & DevOps	AWS (SQS, EC2, S3, IAM, VPC, ELB, DynamoDB) · Azure · Docker · Kubernetes · Ansible · Jenkins · CI/CD · Terraform · Linux
Architecture	Microservices · Event-Driven Design · Rate Limiting · API Gateway · LLD/HLD · Distributed Systems
AI & Dev Tools	Cursor · Claude · GitHub Copilot · AI-assisted development · Insomnia · Postman · Draw.io

KEY PROJECTS (OPEN SOURCE)

Kube-Pulse (Active) | Go · Rust · gRPC · Kubernetes
Microservices-based Kubernetes observability system: gateway (Go/REST), collector (Rust/K8s metrics), and analyser (Rust/anomaly detection). Services communicate over gRPC.

CODING PROFILES

Exercism	exercism.org/profiles/krish8learn — Practice tracks across multiple languages
HackerRank	hackerrank.com/profile/krishkarmakar17 — SQL and algorithmic problem sets
Portfolio	krish8learn.github.io — Maintained a personal portfolio to showcase the engineering skills

EDUCATION

B.Tech — Electronics and Communication Engineering

Kalyani Government Engineering College, MAKAUT University · CGPA: 8.5 · 2017–2020

Diploma — Electronics and Telecommunication Engineering

Jnan Chandra Ghosh Polytechnic, West Bengal State Council of Technical Education · CGPA: 8.6 · 2014–2017

CERTIFICATIONS

- Executive PG Certification — Cloud Computing and DevOps | IIT Roorkee / Intellipaat · Mar 2026
- Rust Fundamentals | Duke University / Coursera · Apr 2026
- Golang Master Certification | Udemy · Jan 2021
- PostgreSQL Certification | Udemy · Feb 2021
- Python Certification | Coursera / University of Michigan · Sep 2020
- MongoDB Basic Course | MongoDB University · Mar 2022

LANGUAGES

Bengali (Native) · English (Fluent) · Hindi (Fluent) · German (Basic / A2 — actively learning)